

Automatic Identification System (AIS): Sampling, Vessel Behaviour, and Traffic Density Analysis

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1 Introduction

The Automatic Identification System (AIS) is a maritime tracking technology that requires vessels above a certain tonnage to continuously broadcast identifying and positional information over VHF radio (International Maritime Organization, 2024). Each AIS message falls into one of two broad categories: *static* information, which changes rarely and includes attributes such as the vessel’s name, call sign, flag state, and physical dimensions; and *dynamic* information, which is updated every few seconds to minutes and includes the vessel’s position, speed over ground, and course over ground. AIS was originally introduced for collision avoidance, but the resulting data stream has since become a valuable resource for studying global shipping patterns, port activity, and inland waterway traffic.

This report analyses an AIS dataset provided through a PostgreSQL/PostgREST backend maintained by the AIDAHO certificate program. We first provide a descriptive overview of the database (Section 2–3), then construct and compare two different sampling strategies for the high-frequency dynamic AIS table (Section 4), examine the movement behaviour of individual vessels including an automated ship-lock detection procedure (Section 5), and – where completed – relate AIS traffic to the German river network (Section 6). Finally, Section 7 documents the deployed static dashboard and Shiny application.

2 Data

2.1 Data Sources and Structure

The database exposes (at least) two tables relevant to this report through a PostgREST API¹:

- **ais_static**: one row per vessel (**mmsi**), containing the vessel name, call sign, flag, ship type, and physical dimensions (length, width, draught).
- **ais_dynamic**: one row per AIS position broadcast, implemented as a TimescaleDB hypertable partitioned by **msg_timestamp**, containing coordinates, speed over ground, course over ground, heading, and a PostGIS **geom** column derived automatically from latitude and longitude.

Both tables are linked via the Maritime Mobile Service Identity (**mmsi**), a unique identifier assigned to each vessel’s AIS transponder.

2.2 Coverage and Limitations

AIS coverage is not uniform. Terrestrial base stations can only receive transmissions within line-of-sight of the coast (typically up to 40–70 nautical miles), so vessels far from shore are only visible through satellite-based AIS receivers, which sample the radio spectrum less frequently and with coarser temporal resolution. As a result, regions and vessel types with predominantly coastal or inland traffic are likely over-represented relative to deep-sea shipping lanes – a form of *availability bias* that is revisited in Section 4.

The data also contains data-quality artefacts. For example, MMSI 2579999 – inspected in Section 5 – does not correspond to a nine-digit, ITU-compliant vessel identifier. Public AIS lookup services list this MMSI as an unnamed base station rather than a vessel, and it additionally appears on a public list of MMSIs associated with anomalous or spoofed AIS traffic. This illustrates that not every MMSI in the database corresponds to a genuine, uniquely identified vessel, and that downstream analyses should treat very short or non-standard MMSIs with caution.

¹Endpoint: <https://aidaho-edu.uni-hohenheim.de/aisdb/>

3 Descriptive Overview

3.1 Available API resources

Querying the PostgREST root endpoint lists the following resources available for analysis: `ais_static`, `ais_dynamic`.

3.2 Database size and vessel counts

Table 1 summarises the size of the two main tables. Counts were obtained via PostgREST's server-side `count()` aggregate, so no raw rows needed to be downloaded into R.

Table 1: Database overview. Values obtained via `02_overview.R`.

Data resource	Number of rows	Number of distinct vessels
Static vessel metadata (<code>ais_static</code>)	[TODO: n]	[TODO: n]
Dynamic AIS position messages (<code>ais_dynamic</code>)	[TODO: n]	[TODO: n]

3.3 Vessel types and flags

The most frequent vessel types in `ais_static` are **[TODO: top 2-3 ship_type values with counts]**. Cargo vessels are expected to dominate the dataset, since AIS carriage is mandatory primarily for commercial ships above a certain gross tonnage, while small leisure craft are typically exempt; this regulatory threshold mechanically shifts the recorded population toward larger commercial vessel classes.

Vessels in the dataset operate under **[TODO: n]** distinct flags. The most common flag is **[TODO: flag code]** with **[TODO: n]** vessels, which is consistent with the dominance of a small number of large open- and traditional-registry states in global merchant shipping.

3.4 Column overview of `ais_static`

Table 2 reports summary statistics for the numeric columns of `ais_static`. A value of exactly 0 for `draught`, `length`, or `width` is treated as "unknown" rather than a genuine physical measurement of zero, and excluded from the non-missing count accordingly.

Table 2: Summary statistics for numeric columns in `ais_static`.

Column	N (valid)	Min	Max	Mean	N (missing)
Draught (m)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Length (m)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Width (m)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

3.5 Filtered queries

Using combinations of PostgREST URL filters (documented in full in `02_overview.R`), we find:

- **Benelux-flagged vessels** (`flag=in.(BE,NL,LU)`): **[TODO: n]** vessels.
- **German cargo vessels** (`flag=eq.DE&ship_type=eq.Cargo`): **[TODO: n]** vessels, of which **[TODO: n]** exceed 150 m in length (`length=gt.150`).
- **Vessels with "EXPRESS" in their name** (`name=ilike.*EXPRESS*`): **[TODO: n]** vessels.

- **Dynamic records in the first five minutes of 2022** (`msg_timestamp=gte.2022-01-01T00:00:00Z&msg` **[TODO: n]** records from **[TODO: n]** distinct vessels.
- **Fast-moving vessels on 2021-05-04, 14:00–14:30 UTC** (`speed>gt.12` within the time window): **[TODO: n]** vessels exceeded 12 knots, of which **[TODO: n]** were classified as cargo vessels.

The `limit` parameter caps the number of rows returned by a single request (e.g. `GET /ais_static?limit=5`); this was used throughout to keep exploratory requests against `ais_dynamic` inexpensive for the shared server.

4 Sampling Strategy

Because `ais_dynamic` contains hundreds of millions of rows, an exploratory analysis cannot operate on the full table. A *sampling strategy* is a deliberate, documented procedure for selecting a subset of observations from a larger population such that the subset can support valid inference (or, at minimum, an informative description) about that population. Sampling is not merely "using less data": an arbitrary or convenient subset – for example, the first n rows returned by a database, or all rows from a single hour – can systematically misrepresent the underlying population. A good sample should approximate the relevant distributional properties of the population (e.g. the mix of vessel types, geographic spread, and time-of-day patterns) closely enough that conclusions drawn from the sample generalise.

A sample is *representative* along a given dimension if its distribution on that dimension matches the population's distribution. Three dimensions along which an AIS sample could fail to be representative are:

1. **Temporal**: sampling only daytime hours would over-represent vessel types and traffic patterns specific to that period (e.g. ferry schedules) and miss night-time patterns.
2. **Vessel type**: cluster sampling by time interval implicitly over-samples vessel types that transmit more frequently or are more numerous at a given moment (e.g. cargo vessels in a busy shipping lane), under-representing rarer types.
3. **Geographic / receiver-type**: coastal, terrestrial-receiver areas are sampled far more densely than open ocean, which is only visible via satellite AIS.

Availability bias occurs when the ease of observing a unit is correlated with some property of that unit, so that the observed sample systematically over- or under-represents units that are harder to detect. In AIS, terrestrial receivers can only detect vessels within line-of-sight of the coast, while satellite receivers cover the open ocean but at a lower temporal resolution. As a result, near-shore and inland-waterway vessel activity is over-represented relative to its true share of global vessel-time, while open-ocean transits – particularly of vessels that do not pass through satellite re-broadcast windows – are under-represented. This means studies that rely only on terrestrial AIS will systematically miss deep-sea fishing and long-haul container traffic far from shore.

4.1 Interval-Based Sample

Sampling frame. All 288 five-minute intervals covering 2024-01-24 form the sampling frame. **Procedure.** 100 intervals were drawn uniformly at random without replacement; up to 100 observations were retrieved from each selected interval, stopping once 10,000 observations had accumulated in total. **Sample size.** **[TODO: n]** observations across **[TODO: n]** intervals were ultimately retrieved (see `01_data/sample_intervals.csv`). **Reproducibility.** A fixed seed (`set.seed(2026)`) was used. This matters because both the choice of intervals and any later random draws build on R's pseudo-random number generator: without a fixed seed, every re-run

of the script – including a grader’s re-run – would draw a different sample, making it impossible to reproduce the exact figures, tables, and downstream comparisons reported here.

This design is a form of *cluster sampling*, where each five-minute interval is a cluster. Two biases this introduces:

1. **Within-cluster homogeneity:** observations within the same five-minute window tend to come from a similar mix of vessels (those active at that moment), so 100 observations from one interval provide less independent information than 100 observations drawn from 100 different moments in time.
2. **Unequal inclusion probabilities for slow-reporting vessels:** vessels that broadcast infrequently (e.g. slow-moving or anchored vessels with longer AIS reporting intervals) are less likely to have a message fall inside any given five-minute window than fast-moving vessels, biasing the sample toward higher speeds.

An alternative that reduces the first (more serious) bias is to draw a simple random sample of individual AIS messages directly (e.g. via systematic sampling across the full day, or true row-level random sampling if supported by the database), which avoids inducing artificial within-cluster correlation.

4.2 Stratified Sample

Strata. All combinations of `ship_type` and hour-of-day on 2024-01-24 were used to define strata. **Allocation.** The 10,000-record budget was allocated proportionally to each stratum’s share of the day’s total record count, with a floor of one observation for any non-empty stratum (so that very small strata are not silently excluded), followed by a rescaling of the largest strata to keep the total at exactly 10,000. **Within-stratum sampling.** Because PostgreSQL does not expose a server-side "random row" operator, each stratum was over-sampled (up to a cap of 500 rows per stratum) and the allocated number of rows was then drawn as a random subsample in R under the fixed seed – a documented approximation to true random retrieval at the database level.

4.3 Comparison

Figures 1–3 compare the two samples along `ship_type`, `speed`, and `collection_type`.

[TODO: 3-5 sentences interpreting the actual differences you observe between the two samples, and linking each difference back to how that sample was constructed – e.g. "the interval sample shows a heavier right tail in speed because five-minute windows happened to coincide with several fast-moving vessels in open water, whereas the stratified design balances this out by allocating capacity across all hours equally."]

5 Vessel Behaviour

5.1 Sample Dashboard

The 10,000-point interval sample was visualised on an interactive `leaflet` map (Figure 4), with point colour encoding vessel speed on a continuous (viridis) colour scale: darker (purple) points indicate near-stationary vessels, while brighter (yellow) points indicate higher speeds. Restricting the map to `collection_type = "satellite"` observations only [TODO: describe whether the spatial pattern becomes sparser / shifts further offshore, consistent with satellite AIS covering open water that terrestrial receivers cannot reach].



Figure 1: Ship type distribution: interval sample vs. stratified sample. The stratified sample is constructed to match the population’s ship-type proportions by design, while the interval sample reflects whichever vessels happened to be active during the 100 randomly drawn five-minute windows.

5.2 Individual Vessel Tracks

MMSI 2579999. As discussed in Section 2.2, this MMSI is too short to be a valid nine-digit vessel identifier and is publicly listed as an unnamed base station rather than a ship, and separately appears on a public spoofing-anomaly MMSI list. The associated dynamic records should therefore be treated as a data-quality artefact rather than genuine vessel movement.

MMSI 563040400. Over the queried window, **[TODO: 2-4 sentences describing the observed trajectory – general route, typical speed, any stops or direction changes]**.

5.3 Ship-Lock Events

A ship lock is expected to leave a distinctive signature in AIS data: speed drops to near zero for an extended period while the vessel is raised or lowered, the recorded position stays essentially fixed, the stop typically lasts on the order of 10–40 minutes, and the local density of AIS observations is often elevated, since AIS reporting intervals shorten as a vessel’s speed and course-change rate decrease.

We implemented a rule-based detection procedure (`detect_lock_events()` in `ais_dynamic_individual_pat`) summarised by the following decision sequence:

1. Sort observations chronologically.
2. Flag a point as a *candidate* if speed < 1 knot.
3. Merge consecutive candidates into one event if the time gap to the previous candidate is ≤ 15 minutes *and* the spatial distance is ≤ 300 m.
4. Discard events shorter than 5 minutes.
5. For each remaining event, report start time, end time, duration, mean location, and the number of observations.

The 1-knot threshold reflects that vessels under power on open water rarely sustain such low speeds, while a genuine lock passage requires coming to a near stop; the 300 m / 15-minute



Figure 2: Speed distribution: interval sample vs. stratified sample.

joint condition keeps a single lock chamber together while preventing two unrelated stops (e.g. a lock stop and a much later anchoring event) from being merged into one event; the 5-minute minimum duration filters out brief manoeuvring slow-downs that are not full lock passages.

Applying this procedure to MMSI 211430830 over 2021-01-17 to 2021-01-20 (Vienna–Linz) yields the events in Table 3.

Table 3: Detected lock passages for MMSI 211430830. Values obtained via `ais_dynamic_individual_paths.R`.

Start time	Duration (min)	Latitude	Longitude	N obs.
[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Limitations. Fixed thresholds are not adaptive across vessel types; GPS/AIS position noise near lock infrastructure can occasionally split one real event into two; and the heuristic cannot distinguish a lock stop from other reasons for stopping (bridge waits, anchoring, technical faults) without external knowledge of lock coordinates. An unsupervised alternative – for example clustering jointly on speed, location, and time (e.g. DBSCAN), or a Hidden Markov Model with a latent "stopped" state – could detect low-speed regimes without manually chosen thresholds and could be validated against publicly available coordinates of known Danube locks.

6 AIS Traffic Density on German Rivers

Due to the time constraints of this submission, the H3-based river-matching analysis (Task 5) was not completed. The general approach would combine `rnaturalearth` river geometries with an H3 hexagonal grid index: H3 indices act as keys into a hash table, so that determining whether an AIS point lies near a river reduces to a constant-time hash lookup rather than a pairwise distance computation between every AIS point and every river segment – an approach that scales far better to the millions of records in `ais_germany`. We focused the available time on Tasks 1–4 and 6–7, which are documented in full above and in Section 7.



Figure 3: Collection type distribution: interval sample vs. stratified sample.

Figure 4: Sample dashboard: AIS positions coloured by speed (knots). Screenshot of the interactive map produced by `sample_dashboard.R`.

7 Dashboard and Deployment Evidence

Table 4: Dashboard and deployment evidence.

Component	Evidence
Static map dashboard	[TODO: URL or status]
Interactive Shiny application	[TODO: URL or status]

The static dashboard (Section 6 of the assignment) reuses the speed-coloured leaflet map from Section 5, exported as a self-contained HTML file via `htmlwidgets::saveWidget()` and served directly by an NGINX container. The Shiny application (Section 7, Option B) reads a pre-aggregated CSV (produced once by `shiny_data_prep.R`, rather than querying the API live on every interaction) and lets the user select an hour, a collection type, and one or more ship types; after clicking "Refresh plot", it displays the number of AIS records per minute (0–59) for the selected filters. NGINX reverse-proxies requests to `/ais_app` to the Shiny container, while `/sample_points.html` is served as a static file from the same NGINX instance. **[TODO: 2-4 sentences documenting your actual deployment: did both components come up successfully? Any debugging needed (ports, volume mounts, NGINX location blocks)? Include a screenshot reference if you took one.]**

8 Conclusion

[TODO: final conclusion paragraph]

Figure 5: Trajectory of MMSI 211430830 with detected ship-lock events highlighted in red.

9 Declaration on the Use of AI Tools

We used Claude (Anthropic) to support the structuring of the R scripts (API helper functions, sampling logic, and the rule-based ship-lock detection function), to draft the LaTeX report skeleton, and to help interpret the PostgREST documentation. All numerical results, figures, and tables in this report were generated by running the submitted R scripts against the live AIS database, and were checked by us against the underlying data before inclusion. We did not use AI-generated citations without verifying the original sources, and we are able to explain and reproduce every part of this submission.

References

International Maritime Organization (2024). *Automatic Identification Systems: Overview and Background*. <https://example.org/ais-overview>. Placeholder reference. Replace with the source actually used.

A Appendix: Reproducibility and Repository Information

A.1 Repository Structure

- 00_docs: assignment sheet and supporting documents.
- 01_data: generated sample data (`sample_intervals.csv`, `sample_stratified.csv`) and intermediate data products.
- 02_code: analysis scripts (R/) and reusable helper functions (`functions/api_helpers.R`).
- 03_report: this report and all figures (`graphs/`).

A.2 Execution Order and Reproduction Notes

Scripts are run from the project root in the following order: `02_overview.R` → `ais_dynamic_sample_points.R` → `sample_dashboard.R` → `ais_dynamic_individual_paths.R` → `sample_html_dashboard.R` → `shiny_data_prep.R`. Full details are documented in `README.md`.

A.3 Software Environment

Table 5: Software environment used to produce the submitted results.

Component	Version or description
R version	[TODO: output of <code>R.version.string</code>]
Operating system	[TODO: e.g. Windows 11 / macOS / Ubuntu]
Package management	[TODO: e.g. <code>renv</code> lockfile available: yes/no]

A.4 Important Submitted Files

- `README.md`: instructions for reproducing the analysis.
- `01_data/sample_intervals.csv`: interval (cluster) sample.
- `01_data/sample_stratified.csv`: stratified sample.
- `03_report/graphs/`: all figures used in this report.